Task : SOC Fundamentals and Security Operations Activity

1. Objective

The objective of this task is to understand SOC fundamentals and security operations by learning theoretical concepts of security monitoring, log management and incident response, and implementing practical tasks including log collection pipeline setup, vulnerability scanning using OpenVAS, intrusion detection using Snort, and monitoring dashboard configuration using ELK Stack in a virtualized lab environment.

2. Tools Used

The following tools were used to complete this task:

- Oracle VirtualBox and VMware for virtualization
- Kali Linux as primary security testing machine
- Metasploitable 2 as vulnerable target system
- ELK Stack (Elasticsearch, Logstash, Kibana) for SIEM and log management
- OpenVAS for vulnerability scanning
- Snort for intrusion detection
- Wireshark for network traffic analysis

3. SOC Fundamentals and Operations

3.1 Purpose of SOC

A Security Operations Center serves three main purposes in an organization:

- Proactive threat detection to identify threats before damage occurs
- Incident response to handle security events effectively
- Continuous monitoring of network and systems around the clock

3.2 SOC Team Roles

The SOC team consists of analysts at different levels:

- Tier 1 Analyst handles initial alert monitoring and basic triage
- Tier 2 Analyst performs deeper investigation of escalated incidents
- Tier 3 Analyst deals with advanced threats and threat hunting activities
- SOC Manager coordinates team activities and reports to management

3.3 Key Functions

Wireshark The main functions performed in a SOC include:

- Log analysis to identify suspicious activities in system and network logs
- Alert triage to prioritize which security alerts need immediate attention
- Threat intelligence integration to stay updated on latest attack techniques

3.4 Frameworks Studied

Two important frameworks were studied for SOC operations:

- NIST Cybersecurity Framework provides guidelines for security program
- MITRE ATT&CK Framework maps attacker techniques and tactics

4. Security Monitoring Basics

4.1 Objectives of Security Monitoring

Security monitoring aims to achieve the following:

- Detect anomalies in network traffic and user behavior patterns
- Identify unauthorized access attempts to systems and data
- Find policy violations by users or applications

4.2 Monitoring Tools

Different categories of tools are used for security monitoring:

- SIEM tools like Elastic SIEM for centralized log collection and analysis
- Network analyzers like Wireshark for packet level inspection
- Vulnerability scanners like OpenVAS for finding system weaknesses

4.3 Key Metrics

Important metrics used in security monitoring include:

- False Positive means alert triggered when no actual threat exists
- False Negative means real threat was missed by detection system
- MTTD (Mean Time to Detect) is average time taken to identify a threat

4.4 ELK Stack Components

The ELK Stack was used for SIEM implementation:

- Elasticsearch stores and indexes all collected log data
- Logstash collects and processes logs from different sources
- Kibana provides web dashboard for visualization and analysis

Figure 1: ELK Stack Services Status

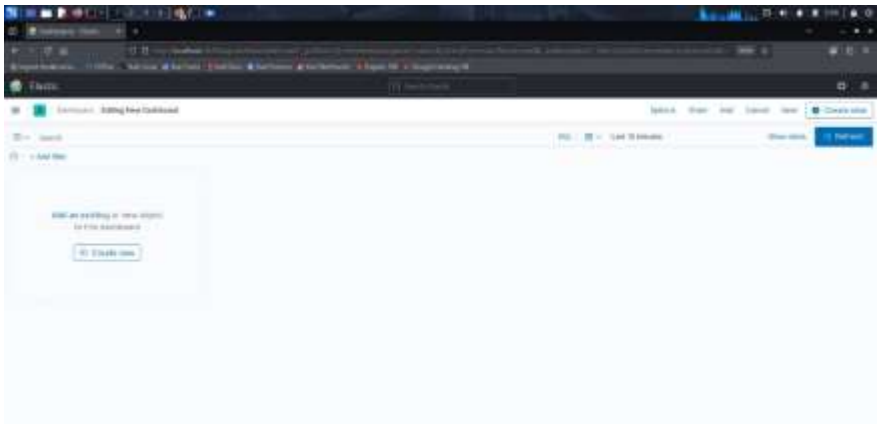

Figure 2: Kibana Dashboard Interface

5. Log Management Fundamentals

5.1 Log Lifecycle

Log management follows a specific lifecycle:

- Collection involves gathering logs from different sources
- Normalization converts logs to standard format for analysis
- Storage keeps logs in centralized location
- Retention defines how long logs are kept
- Analysis examines logs to find security issues

5.2 Common Log Types

Different types of logs are analyzed in SOC:

- Windows Event Logs contain security and system events
- Syslog is standard logging format for Linux and network devices
- HTTP server logs record web server access and errors

5.3 Log Collection Pipeline Setup

A log collection pipeline was configured using Logstash to collect and forward logs to Elasticsearch for analysis in Kibana.

The following configuration was created:

```
input {
file {
path => "/var/log/syslog"
start_position => "beginning" } }
output {
elasticsearch {
hosts => ["localhost:9200"]
index => "syslog-%{+YYYY.MM.dd}" } }
```

Figure 3: Logstash Configuration File

5.4 Log Generation and Testing

Test logs were generated using logger command to verify the pipeline was working correctly. The logs appeared in Kibana Discover section after few seconds.

Figure 4: Test Log Generation

Figure 5: Logs Received in Kibana Discover

6. Security Tools Implementation

6.1 Tools Categories

Security tools are categorized based on their function:

- SIEM tools like Splunk and QRadar for log correlation
- EDR tools like CrowdStrike for endpoint detection
- IDS/IPS tools like Snort for intrusion detection
- Vulnerability scanners like Nessus and OpenVAS

6.2 Snort IDS Setup

Snort was installed on Kali Linux to detect malicious network activity. A custom rule was created to detect requests to suspicious domains.

The following rule was added to local.rules file:

alert tcp any any -> any 80 (msg:"Malicious Domain Access Detected"; content:"malicious.com"; sid:1000001; rev:1;)

alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000002; rev:1;)

Figure 6: Snort Custom Rules

Snort was started in console mode and tested by generating ICMP traffic using ping command. The alert was successfully generated when ping packets were detected.

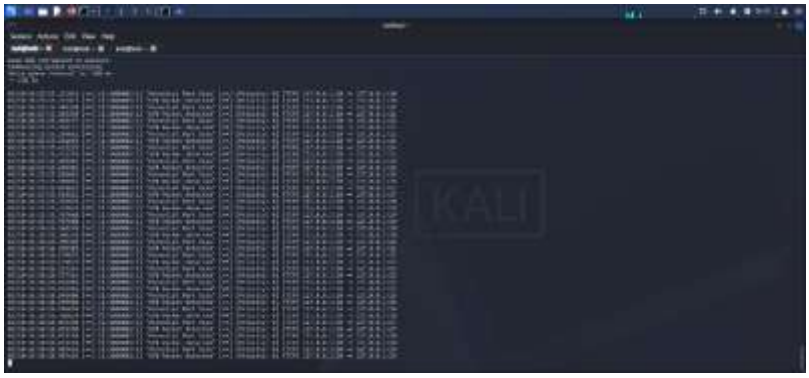

Figure 7: Snort Alert Detection

6.3 Vulnerability Scanning with OpenVAS

OpenVAS was configured to scan the Metasploitable 2 target machine. A full vulnerability scan was performed and the results were analyzed.

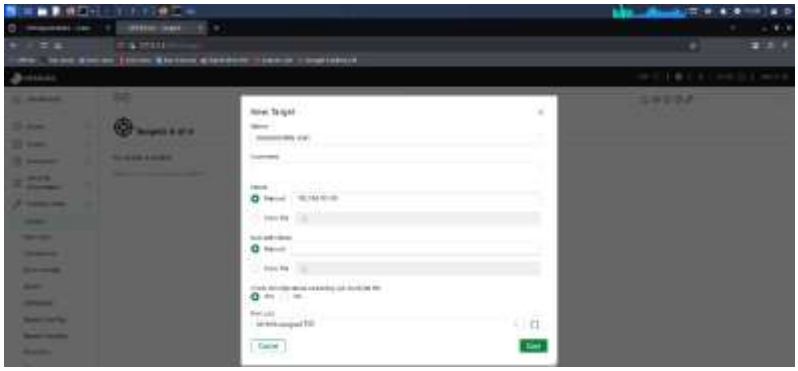

Figure 8: OpenVAS Target Configuration

6.4 Top Vulnerabilities Found

The scan identified several critical vulnerabilities in Metasploitable 2:

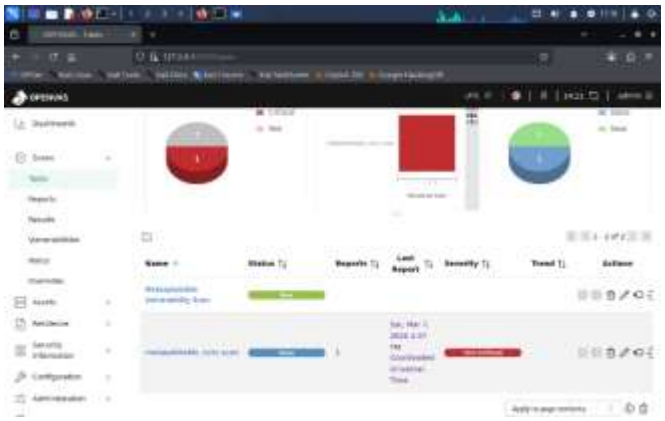

Figure 9: Scan in Progress

S.No	Vulnerability Name	CVSS Score	Severity
1	VsFTPd Backdoor Command Execution	10.0	Critical
2	UnrealIRCd Backdoor Detection	10.0	Critical
3	Samba MS-RPC Remote Shell	9.3	Critical

7. Basic Security Concepts

7.1 CIA Triad

The CIA triad forms the foundation of information security:

- Confidentiality ensures only authorized users access sensitive data
- Integrity ensures data is not modified by unauthorized parties
- Availability ensures systems and data are accessible when needed

7.2 Key Definitions

Important security terminology includes:

- Threat is any potential danger that can exploit a weakness
- Vulnerability is a weakness in system that can be exploited
- Risk is the probability of threat exploiting vulnerability causing damage

7.3 Security Principles

Modern security follows these principles:

- Defense in Depth uses multiple layers of security controls
- Zero Trust assumes no user or system should be trusted by default

7.4 Case Study: Equifax Breach

The 2017 Equifax breach was studied as an example:

- Attackers exploited unpatched Apache Struts vulnerability
- 147 million customer records were stolen
- Root cause was failure to apply security patches on time
- Lesson learned is importance of regular vulnerability management

8. Security Operations Workflow

8.1 Workflow Stages

SOC follows a structured workflow for handling security events:

- Detection stage where alerts are generated from SIEM or EDR tools
- Triage stage where analysts prioritize alerts by severity
- Investigation stage where logs are correlated and IOCs are analyzed
- Response stage where threats are contained and removed

8.2 Phishing Incident Workflow

A workflow was created for handling phishing email incidents:

- User reports suspicious email to security team
- Analyst examines email headers and sender information
- Attachments are analyzed in sandbox environment
- If malicious, sender is blocked and users are notified
- Incident is documented for future reference

9. Incident Response Basics

9.1 IR Lifecycle

The NIST SP 800-61 framework defines six phases of incident response:

- Preparation involves setting up tools, team and procedures
- Identification is detecting that an incident has occurred
- Containment stops the incident from spreading further
- Eradication removes the threat from the environment
- Recovery restores systems to normal operation
- Lessons Learned documents findings and improves processes

9.2 Tabletop Exercise

A ransomware attack scenario was simulated as tabletop exercise:

- Scenario: Ransomware detected on employee workstation
- Step 1: Isolate affected machine from network immediately
- Step 2: Identify ransomware variant using file signatures
- Step 3: Check if clean backups are available
- Step 4: Scan other systems for infection indicators
- Step 5: Restore affected system from clean backup
- Step 6: Update security controls to prevent recurrence

10. Documentation Standards

10.1 Documentation Types

SOC uses various documentation for operations:

- Incident Reports record details of security events
- Runbooks provide step by step procedures for common tasks
- SOPs define standard operating procedures for team
- Post-mortems analyze incidents after resolution

10.2 Security Event Documentation Template

A template was created for documenting security events:

Date/Time	Source IP	Event ID	Description	Action Taken
[Date] 10:15	192.168.1.45	4625	Multiple failed login attempts	Monitored, blocked after 10 attempts

[Date] 11:30	10.0.0.25	7045	New service installed	Investigated, confirmed legitimate
[Date] 14:45	192.168.1.100	4720	New user account created	Verified with IT admin

Figure 10: Security Event Documentation Template

11. Log Analysis Practice

11.1 Syslog Analysis

System logs were analyzed on Kali Linux to identify security events. The syslog file was examined for authentication related messages. The following command was used to filter security events:
>>cat /var/log/syslog | grep -i "auth\|fail\|error" | tail -20

11.2 Failed Login Detection

Failed SSH login attempts were generated and analyzed. Multiple login attempts with wrong password were made and the corresponding log entries were captured.

Figure 11: Failed SSH Login Attempts

12. Monitoring Dashboard Setup

12.1 Kibana Dashboard Configuration

A monitoring dashboard was created in Kibana to visualize security events. The dashboard included visualizations for top source IPs and event frequency.

12.2 Visualizations Created

The following visualizations were added to the dashboard:

- Bar chart showing top 10 source IPs by event count
- Line chart showing event frequency over time
- Data table showing recent security events

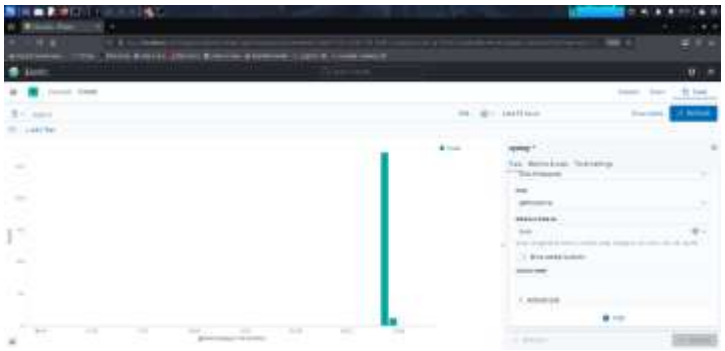

Figure 12: Kibana Event Visualization

Figure 13: SOC Monitoring Dashboard

13. Alert Rule Configuration

13.1 Alert Rule Setup

An alert rule was configured to detect multiple failed login attempts within a short time period. This helps identify potential brute force attacks.

13.2 Rule Configuration

The following rule was created:

- Rule Name: Multiple Failed Logins Detection
- Condition: More than 5 failed logins in 5 minutes
- Index Pattern: syslog-*
- Action: Generate alert in dashboard

13.3 Testing

The rule was tested by generating multiple failed SSH login attempts. The alert was triggered successfully when the threshold was exceeded.

14. Challenges Faced

Several challenges were encountered during the task completion:

- Elasticsearch consumed high RAM causing system slowdown. This was resolved by adjusting heap size in elasticsearch.yml configuration to 2GB.
- The vulnerability scan took longer than expected to complete. Partial results were documented and full scan was left running.
- Initial Snort configuration had incorrect network interface. This was fixed by identifying correct interface using ip addr command.
- Some logs were not appearing in Kibana immediately. Logstash was restarted and index pattern was recreated to fix the issue.

15. Key Learnings

The following key learnings were gained from this task:

- SOC operations require combination of people, processes and technology working together
- Log management is foundation of security monitoring as without proper logs threats cannot be detected
- SIEM tools like ELK Stack are powerful for centralizing and correlating security data from multiple sources
- Vulnerability scanning should be done regularly to identify and fix weaknesses before attackers exploit them
- Incident response requires proper planning and documentation is essential for learning from security events
- Alert tuning is critical to reduce false positives and help analysts focus on real threats
- MITRE ATT&CK framework helps understand attacker behavior and what indicators to look for
- Hands-on practice is essential as theoretical knowledge alone is not sufficient for SOC analyst role

16. Conclusion

This task provided comprehensive understanding of SOC fundamentals and security operations. The lab environment using Kali Linux, Metasploitable 2 and ELK Stack enabled practical implementation of concepts learned in theoretical sections.

Key activities completed include log collection pipeline configuration using Logstash, vulnerability scanning using OpenVAS against Metasploitable target, intrusion detection setup using Snort with custom rules, and monitoring dashboard creation in Kibana.

The practical experience gained through this task will be valuable for understanding real world SOC analyst responsibilities including log analysis, alert triage, threat detection and incident documentation.